# FLASQUE MODULES HAVE VANISHING HIGHER CECH COHOMOLOGY FOR EVERY OPEN COVERING

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The chapter text immediately before the lemma states that the result fails for flasque presheaves; that warning lies outside the counted span and is the reason the statement is given for sheaves.

*Editorial note.* The counted spans carry 2 navigation stubs and no bibliography citation.

*Editorial note.* The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

*Editorial note.* The label lemma-flasque-acyclic-cech is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

*Allowed standard prerequisites (editorial).*

- The Cech-to-cohomology spectral sequence and its low degree terms. Same chapter, Lemma lemma-cech-spectral-sequence, invoked at proof lines 1973-1974. Supplies the spectral sequence whose input presheaves are shown to vanish.
- Flasque modules are acyclic for the global section functors. Same chapter, Lemma lemma-flasque-acyclic, invoked at proof lines 1974-1977. Supplies both the vanishing of the higher cohomology presheaves and the vanishing of the cohomology groups.

*Source reference.* Original lemma 01ES: sources/stacks-cohomology/pinned-original-source.tex, source lines 1541–1541. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original lemma 09SY: sources/stacks-cohomology/pinned-original-source.tex, source lines 1913–1913. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

**Lemma 0.1.** *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules. Let $\mathcal{U} : U = \bigcup U_i$ be an open covering. If $\mathcal{F}$ is flasque, then $\check{H}^p(\mathcal{U}, \mathcal{F}) = 0$ for $p > 0$.*

*Proof.* The presheaves $\underline{H}^q(\mathcal{F})$ used in the statement of Lemma 01ES are zero by Lemma 09SY. Hence $\check{H}^p(U, \mathcal{F}) = H^p(U, \mathcal{F}) = 0$ by Lemma 09SY again. □

*Date*: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.